A Predictive Analysis of Amazon Best Sellers

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DATASET & RESEARCH QUESTION

What product characteristics predict Best Seller status on Amazon?

- **Target:** `isBestSeller` (binary) — only **0.41%** of products qualify
- **Continuous predictors:** price, star rating, review count — all right-skewed
- **Categorical predictor:** `category_name` — 250+ levels, high dimensionality
- **Core challenges:** extreme class imbalance; non-linear social-proof effects; sparse category structure

1.4M
RAW OBS.

89,957
AFTER FILTER

0.41%
BS RATE

250+
CATEGORIES

Variable	Median	Mean
Price ($)	26.99	48.90
Stars	4.5	4.39
Reviews	3	501

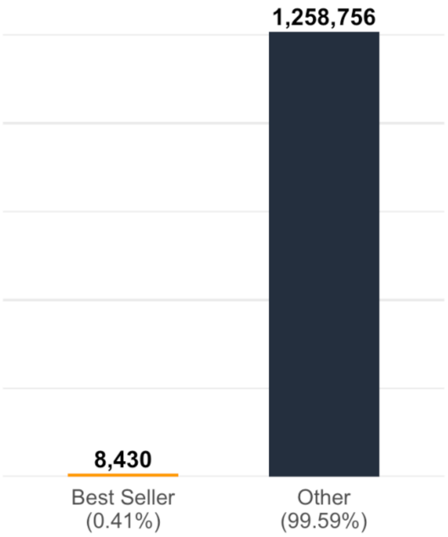
Pre-processing pipeline

- Filter `price > 0 & stars > 0` — removes 93.7% of raw rows
- Log-transform: `log_price = log(price)`, `log_reviews = log1p(reviews)`
- Stratified 70/30 train/test split, `set.seed(1)`

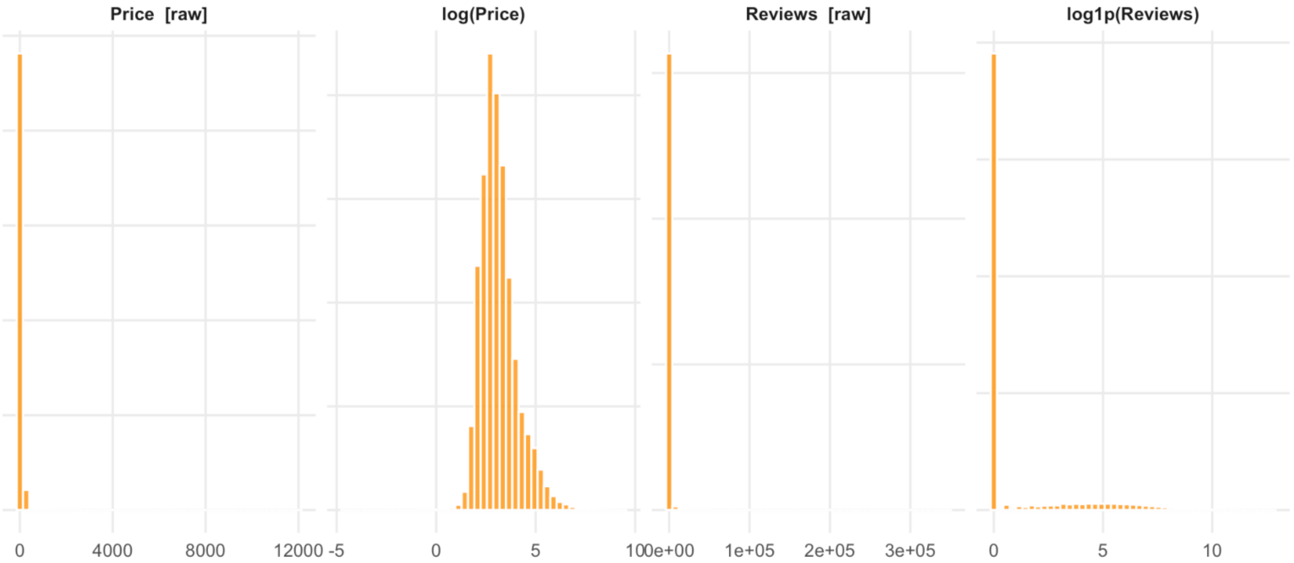
Source: Amazon Products Dataset 2023 — Kaggle

EDA — CLASS IMBALANCE & PREDICTOR SKEW

Class Imbalance

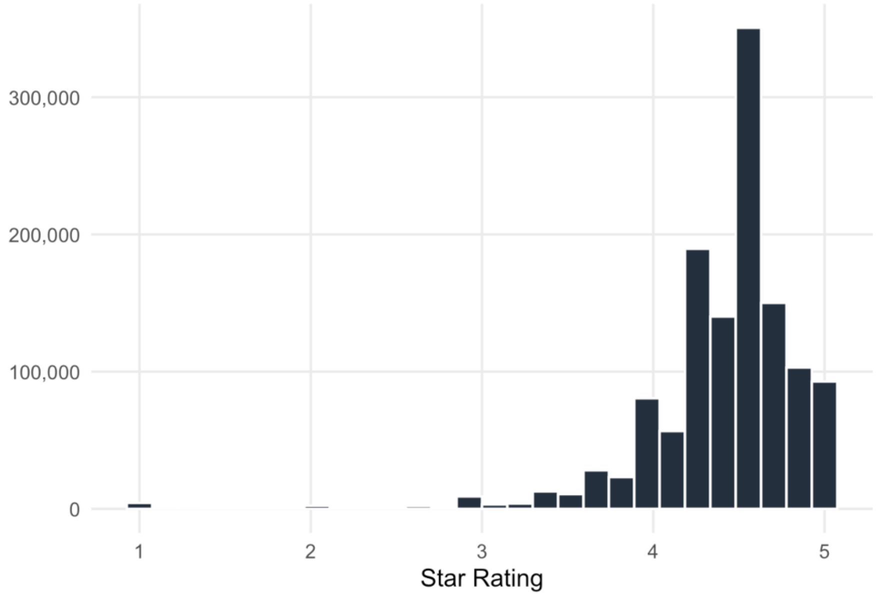


Before and After Log Transform

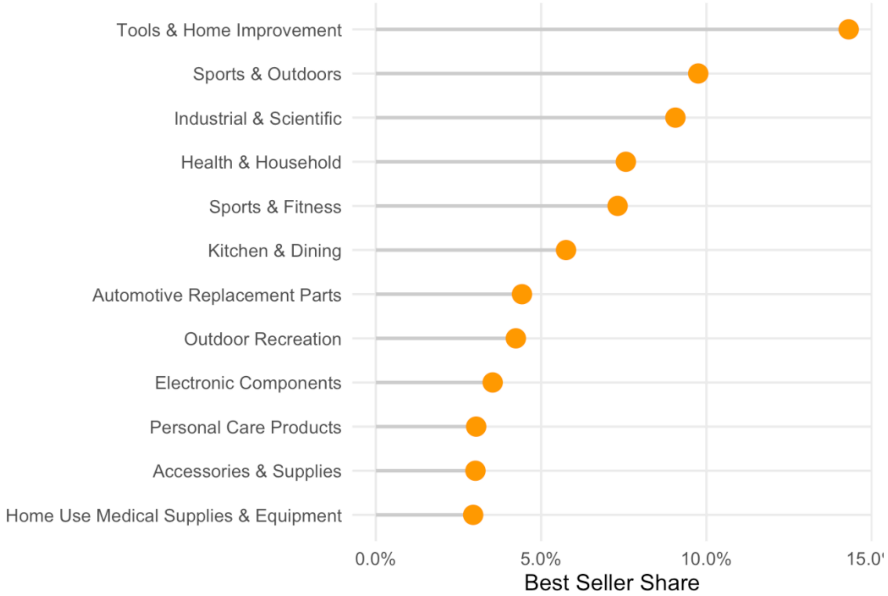


EDA — RATING DISTRIBUTION & CATEGORY PATTERNS

Rating Distribution

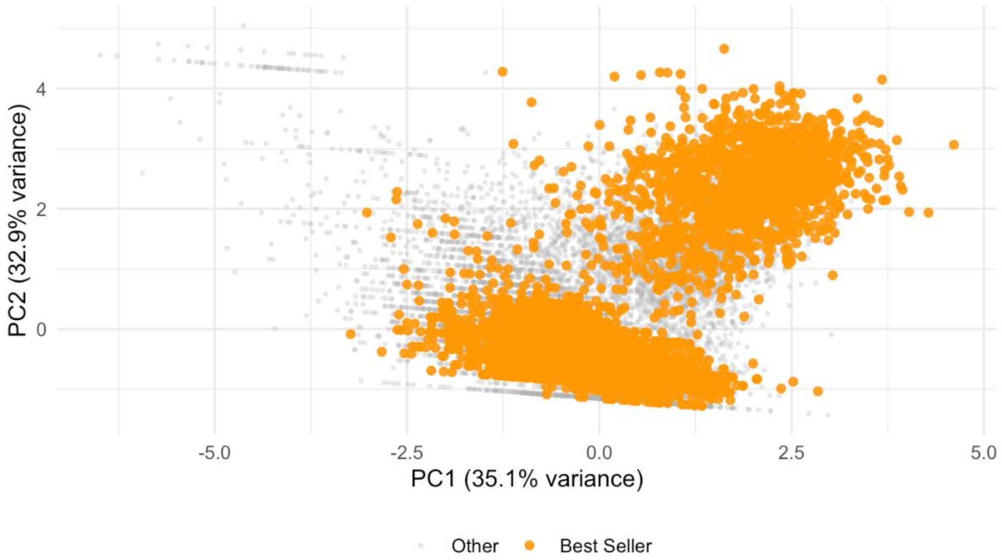


Top 12 Categories by Best Seller Share



UNSUPERVISED EXPLORATION

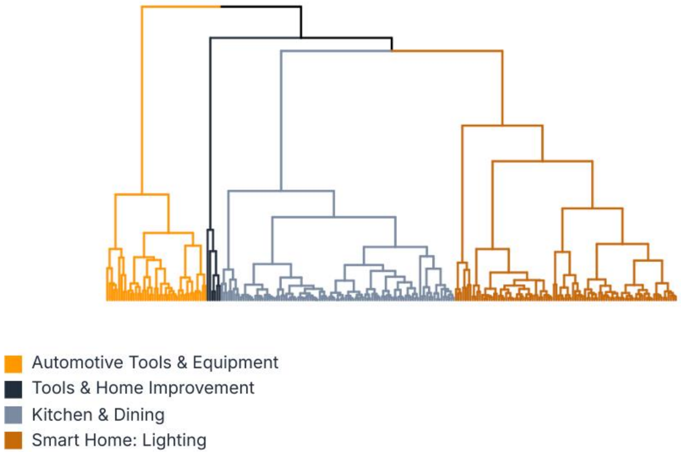
PCA on numeric predictors



Best Sellers **overlap** the negative-class cloud — not linearly separable in numeric space alone.

Category clustering (Ward, k = 4)

- Apparel → **lowest** BS density
- Specialty electronics → **highest** BS density
- Review volume drives cluster separation



METHODOLOGY

🔷 Model 1: Logistic Lasso

- L_1 penalty → **auto-selects** relevant categories from 250+
- 10-fold CV, AUC criterion
- Inverse-prevalence class weights

$$\hat{\beta} = \operatorname{argmin} \{ -(1/n)\Sigma[y_i x_i^T\beta - \log(1+e^{x_i^T\beta})] + \lambda\|\beta\|_1 \}$$

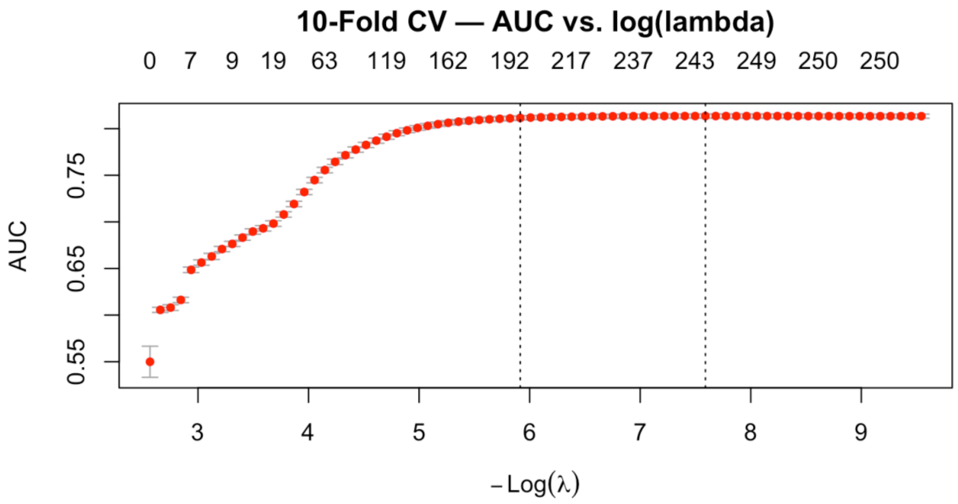
🔶 Model 2: GAM + Smoothing Splines

- Captures **diminishing returns** of review volume
- No pre-specified functional form needed
- REML smoothing parameter selection

$$\operatorname{logit} P(Y=1) = \beta_0 + f_1(\text{log\_reviews}) + f_2(\text{log\_price}) + \beta_1 \cdot \text{stars}$$

Evaluated on the **30% held-out test set** — AUC and PR-AUC only (accuracy is degenerate at 0.41% prevalence).

MODEL 1 — LASSO RESULTS



243
NON-ZERO VARS

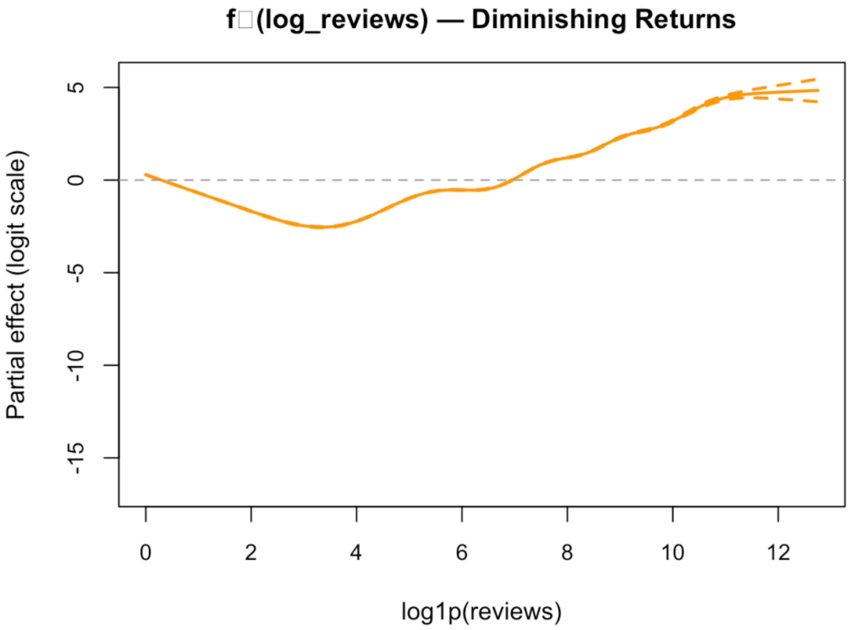
240
CATEGORIES KEPT

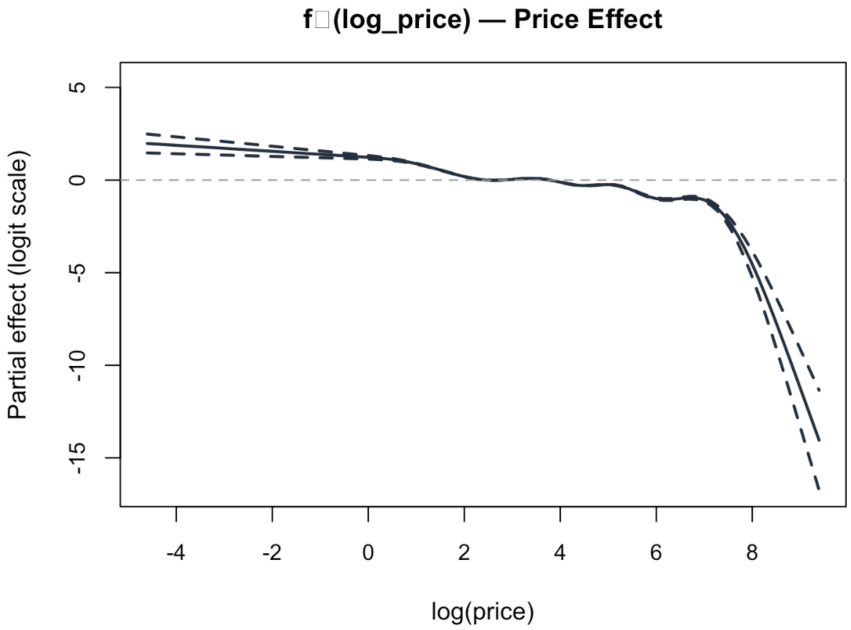
0.814
BEST CV-AUC

Top 8 by |coeff.|

Term	Coeff.
cat: Nintendo Switch Consoles, Games & Accessories	-4.441
cat: Building Toys	-3.927
cat: Online Video Game Services	+3.805
cat: Kids' Play Cars & Race Cars	-3.796
cat: Laptop Bags	-3.756
cat: Headphones & Earbuds	-3.729
cat: GPS & Navigation	-3.715
cat: Tools & Home Improvement	+3.704

MODEL 2 — GAM RESULTS





MODEL COMPARISON & EVALUATION

Model	ROC-AUC	PR-AUC
Lasso	**0.8134**	**0.0435**
GAM	0.6682	0.0258
Baseline (all-zero)	0.5000	0.0067

Lasso

PRIMARY MODEL (PR-AUC)

Why these metrics?

- **Not accuracy** — saying "nothing is a Best Seller" is 99.59% accurate but finds zero. A useless model should not score well.
- **ROC-AUC** — of a random Best Seller and a random non-Best Seller, how often does the model rank the right one higher?
- **PR-AUC** — when the model flags a product, how often is it actually a Best Seller? Harder to fake with rare classes. Random baseline = 0.0067.

> The best model finds Best Sellers **6× more often** than picking products at random.

LIMITATIONS & DISCUSSION

Selection bias

- Filter removes 93.7% of catalog
- Excludes free listings and unrated arrivals
- Results apply to *established, priced* products only

Class imbalance

- 0.41% prevalence → inverse-prevalence weighting required
- GAM splines extrapolate at the high-review tail (sparse positives)

Apparel dominance

- 3 largest categories = apparel, lowest BS density
- Lasso coefficients estimated relative to this apparel baseline

Cross-sectional data

- Single snapshot → cannot separate stock from *flow*
- Best Seller status partly reflects sales velocity, not just product attributes

CONCLUSIONS & FUTURE WORK

Key findings

- GAM splines reveal **diminishing returns** of review volume
- **Reputation** (reviews + stars) outweighs price as the main driver
- Lasso zeroes out most categories → signal concentrated in few levels
- Apparel dominates the sample → bounds category-level interpretation

Future Work

📈 **Time-series panel data — model sales velocity (flow) rather than a cross-sectional stock snapshot**

🔤 **NLP on product titles — sentiment scores, brand signals, and category-aware embeddings to improve discrimination in the long tail of low-review products**